

Homework 1 (1.1.1 ~ 1.4.2)

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1.1.1

a. $|0\rangle$ is uniqueProof: (i) $|V\rangle + |0\rangle = |V\rangle$ Where $|V\rangle$ is an arbitrary state ket.

(ii) $|V\rangle + |0'\rangle = |V\rangle$

Set $|V\rangle = |0'\rangle$ in (i) $\Rightarrow |0'\rangle + |0\rangle = |0'\rangle$

Set $|V\rangle = |0\rangle$ in (ii) $\Rightarrow |0\rangle + |0'\rangle = |0\rangle$

Therefore, $|0'\rangle = |0'\rangle + |0\rangle \xrightarrow{\text{Commutativity of vector addition}} |0\rangle + |0'\rangle = |0\rangle$

b. $0|V\rangle = |0\rangle$ Proof: $1|V\rangle = (1+0)|V\rangle = 1|V\rangle + 0|V\rangle$, where $1|V\rangle = |V\rangle$.

Therefore $|V\rangle = |V\rangle + 0|V\rangle$.

Since $|V\rangle$ is arbitrary here, compared with the definition of $|0\rangle$, which is $|V\rangle + |0\rangle = |V\rangle$ for any $|V\rangle$, plus $|0\rangle$ is unique, we can conclude that $0|V\rangle = |0\rangle$.c. $|-V\rangle = -|V\rangle$ Proof: $|V\rangle + (-|V\rangle) = 0|V\rangle = |0\rangle$ for arbitrary $|V\rangle$. By definition of $|-V\rangle$, which is $|V\rangle + |-V\rangle = 0$, we can conclude that, $|-V\rangle = -|V\rangle$.d. $|-V\rangle$ is the unique additive inverse of $|V\rangle$.Proof: Suppose there exists another vector $|W\rangle$, satisfying $|W\rangle = -|V\rangle$, then $|W\rangle + |V\rangle = -|V\rangle + |V\rangle$

$$= (1-1)|V\rangle$$

$$= 0|V\rangle$$

$$= |0\rangle$$

Add $|-V\rangle$ on both sides, we have

$$|W\rangle + |V\rangle + |-V\rangle = |0\rangle + |-V\rangle$$

LHS = $|W\rangle + (|V\rangle + |-V\rangle) = |W\rangle + |0\rangle = |W\rangle$

RHS = $|-V\rangle$

Therefore, $|W\rangle = -|V\rangle$.

1.1.2

Null vector: By definition $|V\rangle + |0\rangle = |V\rangle$ for any $|V\rangle$.

Set $|0\rangle = (a_0, b_0, c_0)$, $|V\rangle = (a, b, c)$, where a, b, c are arbitrary numbers.

$$\begin{aligned} \text{Then, } |0\rangle + |V\rangle &= (a_0, b_0, c_0) + (a, b, c) \\ &= (a_0 + a, b_0 + b, c_0 + c) \\ &= |V\rangle \\ &= (a, b, c) \end{aligned}$$

$$\Rightarrow \begin{cases} a_0 + a = a \\ b_0 + b = b \\ c_0 + c = c \end{cases} \Rightarrow \begin{cases} a_0 = 0 \\ b_0 = 0 \\ c_0 = 0 \end{cases}$$

Therefore, the null vector is $(0, 0, 0)$.

Inverse vector:

Suppose the inverse vector of (a, b, c) is $(\bar{a}, \bar{b}, \bar{c})$. By definition $(a, b, c) + (\bar{a}, \bar{b}, \bar{c}) = |0\rangle = (0, 0, 0)$.

$$\Rightarrow (a + \bar{a}, b + \bar{b}, c + \bar{c}) = (0, 0, 0)$$

$$\Rightarrow \begin{cases} a + \bar{a} = 0 \\ b + \bar{b} = 0 \\ c + \bar{c} = 0 \end{cases} \Rightarrow \begin{cases} \bar{a} = -a \\ \bar{b} = -b \\ \bar{c} = -c \end{cases}$$

Therefore, the inverse vector of $(a, b, c) = (-a, -b, -c)$.

$\{(a, b, 1)\}$ does not form a vector space because

① It violates the closure under addition, ie.

$$(a_1, b_1, 1) + (a_2, b_2, 1) = (a_1 + a_2, b_1 + b_2, 2) \notin \{(a, b, 1)\};$$

② It violates the closure under scalar multiplication, ie.

$$\lambda(a_1, b_1, 1) = (\lambda a_1, \lambda b_1, \lambda) \notin \{(a, b, 1)\}, \text{ as long as } \lambda \neq 1.$$

③ There is no null vector, ie. $(0, 0, 0) \notin \{(a, b, 1)\}$

④ The inverse does not exist, ie. $(-a, -b, -1) \notin \{(a, b, 1)\}$

If you want to prove the property of closure
in this problem, please mention that $f(x) + g(x)$
and $\alpha f(x)$ are also periodic functions. That is,
 $f(0) + g(0) = f(L) + g(L)$, $\alpha f(0) = \alpha f(L)$.

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1.1.3

- (i) $\{f(x)\}$, $f(0) = f(L) = 0$, form a vector space.
(ii) $\{f(x)\}$, periodic functions obeying $f(0) = f(L)$, form a vector space.
(iii) $\{f(x)\}$, $f(0) = 4$ DO NOT form a vector space;

- ① If $g(x), h(x) \in \{f(x)\}$, then $g(x) + h(x) \notin \{f(x)\}$, since
 $g(0) + h(0) = 8 \neq 4$;
- ② If $g(x) \in \{f(x)\}$, then $\lambda g(x) \notin \{f(x)\}$, since $\lambda g(0) = 4\lambda \neq 4$,
as long as $\lambda \neq 1$;
- ③ No null vector. $g(x) \equiv 0 \notin \{f(x)\}$, since $g(0) = 0 \neq 4$;
- ④ If $g(x) \in \{f(x)\}$, then the inverse $-g(x) \notin \{f(x)\}$, since $-g(0) = -4 \neq 4$.

1.1.4

Suppose $a_1|1\rangle + a_2|2\rangle + a_3|3\rangle = 0$.

We have
$$\begin{pmatrix} 0 \cdot a_1 + 1 \cdot a_2 + (-2) \cdot a_3 & 1 \cdot a_1 + 1 \cdot a_2 + (-1) \cdot a_3 \\ 0 \cdot a_1 + 0 \cdot a_2 + 0 \cdot a_3 & 0 \cdot a_1 + 1 \cdot a_2 + (-2) \cdot a_3 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

$$\Rightarrow \begin{cases} a_2 - 2a_3 = 0 \\ a_1 + a_2 - a_3 = 0 \end{cases}$$

$$\Rightarrow \begin{cases} a_1 = -a_3 \\ a_2 = 2a_3 \end{cases}$$

It is not necessary for a_1, a_2 and a_3 to be 0 together. Therefore,
 $|1\rangle, |2\rangle$ and $|3\rangle$ are linearly DEPENDENT.

1.1.5

Suppose $a_1(1,1,0) + a_2(1,0,1) + a_3(3,2,1) = 0$

Then
$$\begin{cases} a_1 + a_2 + 3a_3 = 0 \\ a_1 + 2a_3 = 0 \\ a_2 + a_3 = 0 \end{cases} \Rightarrow \begin{cases} a_1 = -2a_3 \\ a_2 = -a_3 \end{cases}$$

When $a_3 \neq 0$, a_1, a_2, a_3 can have non-zero values. Therefore,
 $(1,1,0), (1,0,1), (3,2,1)$ are linearly dependent.

Suppose $a_1(1, 1, 0) + a_2(1, 0, 1) + a_3(0, 1, 1) = 0$

Then
$$\begin{cases} a_1 + a_2 = 0 \\ a_1 + a_3 = 0 \\ a_2 + a_3 = 0 \end{cases} \Rightarrow \begin{cases} a_1 = 0 \\ a_2 = 0 \\ a_3 = 0 \end{cases} \text{ is the only solution.}$$

Therefore, $(1, 1, 0)$, $(1, 0, 1)$ and $(0, 1, 1)$ are linearly independent.

1.3.1

Using Gram-Schmidt process here, starting from $\vec{A}$.

$$\vec{e}_1 = \frac{\vec{A}}{|\vec{A}|} = \frac{3\vec{i} + 4\vec{j}}{\sqrt{3^2 + 4^2}} = \frac{3}{5}\vec{i} + \frac{4}{5}\vec{j}$$

$$\begin{aligned} \vec{e}_2' &= \vec{B} - (\vec{B} \cdot \vec{e}_1) \vec{e}_1 \\ &= (2\vec{i} - 6\vec{j}) - \left(\frac{6}{5} - \frac{24}{5}\right) \left(\frac{3}{5}\vec{i} + \frac{4}{5}\vec{j}\right) \\ &= \left(2 + \frac{54}{25}\right)\vec{i} + \left(-6 + \frac{72}{25}\right)\vec{j} \\ &= \frac{104}{25}\vec{i} - \frac{78}{25}\vec{j} \end{aligned}$$

$$\vec{e}_2 = \frac{\vec{e}_2'}{|\vec{e}_2'|} = \frac{4}{5}\vec{i} - \frac{3}{5}\vec{j}$$

Therefore, the new basis is $\vec{e}_1 = \frac{3}{5}\vec{i} + \frac{4}{5}\vec{j}$, $\vec{e}_2 = \frac{4}{5}\vec{i} - \frac{3}{5}\vec{j}$.

1.3.2

Step 1: $|1\rangle = \frac{|I\rangle}{\sqrt{\langle I|I\rangle}} = \frac{1}{3} \begin{pmatrix} 3 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$

Step 2-1: $|2'\rangle = |II\rangle - (\langle 1|II\rangle)|1\rangle$

$$= \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} - 0 \cdot \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

$$= \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}$$

Step 2-2: $|2\rangle = \frac{|2'\rangle}{\sqrt{\langle 2'|2'\rangle}} = \frac{1}{\sqrt{5}} \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 0 \\ \frac{1}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} \end{pmatrix}$

Step 3-1: $|3'\rangle = |\text{III}\rangle - (\langle 1|\text{III}\rangle)|1\rangle - (\langle 2|\text{III}\rangle)|2\rangle$

$$= \begin{pmatrix} 0 \\ 2 \\ 5 \end{pmatrix} - 0 \cdot \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} - \left(0 + \frac{2}{\sqrt{5}} + \frac{10}{\sqrt{5}}\right) \begin{pmatrix} 0 \\ \frac{1}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} \end{pmatrix}$$

$$= \begin{pmatrix} 0 \\ 2 \\ 5 \end{pmatrix} - \begin{pmatrix} 0 \\ \frac{12}{5} \\ \frac{24}{5} \end{pmatrix}$$

$$= \begin{pmatrix} 0 \\ -\frac{2}{5} \\ \frac{1}{5} \end{pmatrix}$$

Step 3-2: $|3\rangle = \frac{|3'\rangle}{\sqrt{\langle 3|3'\rangle}} = \begin{pmatrix} 0 \\ -2/\sqrt{5} \\ 1/\sqrt{5} \end{pmatrix}$

1.3.3

When $|V\rangle = C|W\rangle$

$$\langle V|V\rangle = |C|^2 \langle W|W\rangle = |C|^2 |W|^2$$

$$\langle W|V\rangle \langle V|W\rangle = (C \langle W|W\rangle)(C^* \langle W|W\rangle)$$

$$= |C|^2 \langle W|W\rangle \langle W|W\rangle$$

$$= |C|^2 |W|^4$$

$$\Rightarrow \langle V|V\rangle = \frac{\langle W|V\rangle \langle V|W\rangle}{|W|^2}$$

When two arrows are parallel or anti-parallel with each other, the square of their inner product equals to the product of their norms.

1.3.4

$$\text{Re} \langle V|W\rangle \leq |\langle V|W\rangle| \leq |V||W|$$

$$\Rightarrow 2 \text{Re} \langle V|W\rangle \leq 2 |V||W|$$

Add $|V|^2 + |W|^2$ to both sides of the inequality above, we have

$$\langle V|V\rangle + 2 \text{Re} \langle V|W\rangle + \langle W|W\rangle \leq |V|^2 + |W|^2 + 2 |V||W|$$

$$\text{LHS} = \langle V|V\rangle + \langle V|W\rangle + \langle W|V\rangle + \langle W|W\rangle$$

$$= \langle V+W|V+W\rangle$$

$$= |V+W|^2$$

$$\text{RHS} = (|V| + |W|)^2$$

$$\text{Therefore, } |V+W|^2 \leq (|V| + |W|)^2$$

$$\Rightarrow |V+W| \leq |V| + |W| \quad (\text{the triangular inequality})$$

Attention, we are supposed to prove the equality holds ONLY IF $|V\rangle = a|W\rangle$ where a is a real number.

The equality holds only if the two equalities below hold:

$$\textcircled{1} \quad \langle V|W\rangle = |V||W|$$

$$\textcircled{2} \quad \text{Re} \langle V|W\rangle = |\langle V|W\rangle|$$

From the proof process of the Schwarz inequality in Shankar, we know that equality $\textcircled{1}$ holds only if

$$|Z\rangle = |V\rangle - \frac{\langle W|V\rangle}{|W|^2} |W\rangle = 0$$

which means $|V\rangle$ must be able to be expressed as $\alpha|W\rangle$, where α is a number. To prove that α must be real, we substitute $|V\rangle = \alpha|W\rangle$ into equality $\textcircled{2}$ above.

$$\langle V|W\rangle = \alpha^* \langle V|V\rangle$$

To satisfy equality $\textcircled{2}$, $\langle V|W\rangle$ must be real. Since $\langle V|V\rangle$ is real, α^* must be real. Therefore, α must be a real number.

1.4.1

Given a vector space V^n , one can start with an arbitrary vector $|V\rangle \neq 0$ and construct $n-1$ other vectors orthogonal to this $|V\rangle$ through Gram-Schmidt process. Since these $n-1$ vectors are linearly independent, they span a V^{n-1} subspace. Now we prove that this subspace V^{n-1} is the set of all vectors orthogonal to $|V\rangle$.

$\textcircled{1}$ Every vector in V^{n-1} is orthogonal to $|V\rangle$: Since every vector in V^{n-1} can be expressed as a linear combination of the $n-1$ vectors we constructed above — $|W\rangle = \sum_{i=1}^{n-1} a_i |V_i\rangle$, and $\langle V|W\rangle = \sum_{i=1}^{n-1} a_i \langle V|V_i\rangle = 0$, because $\langle V|V_i\rangle = 0$ for each V_i .

$\textcircled{2}$ Every vector in V^n but outside V^{n-1} is NOT orthogonal to $|V\rangle$. Since this kind of vector can be expressed as $|U\rangle = a|V\rangle + \sum_{i=1}^{n-1} a_i |V_i\rangle$, where $a \neq 0$. Therefore $\langle V|U\rangle = a \langle V|V\rangle = a \neq 0$.

1.4.2

Since $V_1^{n_1}$ and $V_2^{n_2}$ are two subspaces orthogonal to each other, we can take the n_1 basis vectors of $V_1^{n_1}$ and the n_2 basis vectors of $V_2^{n_2}$, and put them together. Because these $n_1 + n_2$ vectors are orthogonal to each other, they can span a $V^{n_1+n_2}$ subspace. We now prove that this $V^{n_1+n_2}$ is nothing but the $V_1 \oplus V_2$.

① Every vector in $V^{n_1+n_2}$ is in $V_1 \oplus V_2$:

Since each vector in $V^{n_1+n_2}$ can be expressed as $|U\rangle = \sum_{i=1}^{n_1} a_i |V_i\rangle + \sum_{j=1}^{n_2} b_j |W_j\rangle$, where $\{|V_i\rangle\}, \{|W_j\rangle\}$ are basis vectors of V_1 and V_2 respectively.

Notice that $\sum_{i=1}^{n_1} a_i |V_i\rangle$ is a vector in V_1 , and $\sum_{j=1}^{n_2} b_j |W_j\rangle$ is a vector in V_2 . Therefore, $|U\rangle$ can be expressed as a combination of vectors from V_1 and V_2 . According to the definition of $V_1 \oplus V_2$ (Definition 12 in Shankar), $|U\rangle \in V_1 \oplus V_2$.

② Every vector in $V_1 \oplus V_2$ is in $V^{n_1+n_2}$.

Every vector in $V_1 \oplus V_2$ can be expressed as $|Z\rangle = C_1 |Z_1\rangle + C_2 |Z_2\rangle$, where $|Z_1\rangle \in V_1^{n_1}$, $|Z_2\rangle \in V_2^{n_2}$. Therefore, $|Z_1\rangle = \sum_{i=1}^{n_1} p_i |V_i\rangle$, and $|Z_2\rangle = \sum_{j=1}^{n_2} q_j |W_j\rangle$. Thus, $|Z\rangle = \sum_{i=1}^{n_1} C_1 p_i |V_i\rangle + \sum_{j=1}^{n_2} C_2 q_j |W_j\rangle$, which lies in $V^{n_1+n_2}$.

Therefore, the dimension of $V_1 \oplus V_2$ is $n_1 + n_2$.